

RESEARCH ARTICLE

Autonomous observations enhance our ability to observe the biological carbon pump across diverse carbon export regimes

Shawnee Traylor ^{1,*} David P. Nicholson ² Samantha J. Clevenger,¹ Ken O. Buesseler,² Eric D'Asaro ³,
Craig M. Lee³

¹The MIT-WHOI Joint Program in Oceanography/Applied Ocean Science and Engineering, Cambridge and Woods Hole, Massachusetts, USA; ²Woods Hole Oceanographic Institution, Woods Hole, Massachusetts, USA; ³Applied Physics Laboratory and School of Oceanography, University of Washington, Seattle, Washington, USA

Abstract

The expansion of autonomous observation platforms offers vast opportunities for analyzing ocean ecosystems and their role in carbon export. As part of the EXport Processes in the Ocean from RemoTe Sensing campaign, we autonomously measured the productivity regimes in two contrasting end-member ecosystem states. The first campaign occurred in the subpolar North Pacific near Ocean Station Papa (Site 1), characterized by iron limitation and a highly regenerative regime. The second captured a springtime bloom in the North Atlantic (Site 2), which typically drives efficient export of productivity. Using a combination of floats and gliders carrying biogeochemical sensors, we quantified gross primary productivity, net community production, and organic carbon export potential (fC_{org}) to assess biological carbon pump strength. Site 2 demonstrated higher cruise-period productivity, with roughly $5\times$ the gross primary productivity and $13\times$ the euphotic zone net community production seen at Site 1. Greater export efficiency at Site 2 was reflected in numerous indices, such as the ratio of new production to net primary productivity (ef -ratio; Site 1: 0.33; Site 2: 0.73), the ratio of sinking particulate organic carbon to net primary productivity (ez -ratio; Site 1: 0.24; Site 2: 0.69), and mean daily fC_{org} (Site 1: 3.4 ± 0.7 ; Site 2: 20.3 ± 2.3 mmol C m⁻² d⁻¹). Together with particulate organic carbon flux derived from thorium-234 measurements, we infer that observed low net community production was almost entirely routed to sinking particulate organic carbon at Site 1, while the much higher net community production at Site 2 resulted in near-equal proportions routed to dissolved organic carbon production and sinking particulate organic carbon.

The ocean plays a critical role in climate regulation through the uptake and storage of carbon dioxide, containing nearly 12 times as much as the terrestrial biosphere, soils, and atmospheric reservoir combined (Ciais et al. 2013). Photosynthesizers in the sunlit surface ocean fix carbon dioxide

(CO₂) into organic carbon, a portion of which is transported to depth via a complex web of ecological mechanisms. The collection of these processes is termed the biological carbon pump (BCP) and transfers carbon through various pathways, including gravitational settling of particulate organic carbon

*Correspondence: straylor@whoi.edu

This is an open access article under the terms of the [Creative Commons Attribution](#) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Editor-in-Chief: K. David Hambright

Data Availability Statement: All data utilized in this study are publicly available on the National Aeronautics and Space Administration's SeaBASS data repository (<https://seabass.gsfc.nasa.gov/experiment/EXPORTS>) as well as the Biological and Chemical Oceanography Data Management Office (<https://www.bco-dmo.org/>). Ancillary data including 6-h barometric pressure, 10-m wind speed and direction from the National Oceanic and

Atmospheric Administration Ocean Station Papa mooring and National Centers for Environmental Prediction reanalysis are available at <https://www.pmel.noaa.gov/co2/story/Papa/> and <https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html>, respectively. Six-hour wind stress fields were accessed from the National Oceanic and Atmospheric Administration Fleet Numerical Meteorology and Oceanography Center (<https://coastwatch.pfeg.noaa.gov/erddap/griddap/erdlasFnTran6.html>). Biogeochemical-Argo data can be found via the United States Global Ocean Data Assimilation Experiment Global Data Assembly Center (<ftp://usgodae.org/pub/outgoing/argo>).

Special Issue: Autonomous Instrumentation and Big Data: New Windows, Knowledge, and Breakthroughs in the Aquatic Sciences. Edited by: Yui Takeshita, Heidi Sosik, Dominique Lefevre, Werner Eckert, Kevin C. Rose and Deputy Editors Julia C. Mullarney, Steeve Comeau, and Elisa Schaum.

(POC) (McCave 1975; Stukel et al. 2018; Boyd et al. 2019), subduction of water masses containing dissolved organic carbon (DOC) (Hansell and Carlson 1998), and biologically mediated processes such as zooplankton migration, respiration, and mortality at depth (Steinberg et al. 2000). The carbon that reaches the deep ocean may remain sequestered from the atmosphere for hundreds to thousands of years. Global estimates of this export flux are comparable to the magnitude of anthropogenic emissions, ranging from 5 to 12 Pg C yr⁻¹ (e.g., Henson et al. 2011; DeVries and Weber 2017). Notably, oceanic CO₂ uptake via air-sea gas exchange accounts for only 2.5 Pg C yr⁻¹, or 20–50% of the magnitude of the BCP, so even minor alterations in the efficiency of the BCP can cause significant changes in the ocean's overall capacity as a carbon sink (Takahashi et al. 2002). For example, Huang et al. (2022) suggested that a 10% reduction in the strength of the BCP could result in a 20–50% reduction in its capacity as a sink.

The potential export of primary production is ultimately reflected in net community production (NCP), equivalent to the residual of gross primary production (GPP) and all respiratory losses. At a steady state, annual NCP represents the maximum amount of organic carbon available for export. The NCP is typically integrated down to a biogeochemically relevant depth horizon, such as the base of the mixed layer (ML) or euphotic zone (EZ) (e.g., Palevsky and Doney 2018; Buesseler et al. 2020a). Despite its importance in evaluating the strength of the BCP, our understanding of global NCP has been historically limited by labor-intensive ship-based measurement techniques (Ducklow and Doney 2013). However, recent approaches leveraging satellites and autonomous assets seek to address these limitations and enhance our understanding of global spatial and temporal variability of the BCP (e.g., Laws et al. 2011; Westberry et al. 2012). Another significant barrier to predicting the global export flux is the difficulty of resolving the vertical transmission of carbon from the surface ocean to depth (Buesseler and Boyd 2009; Burd et al. 2010). While satellite observations offer near-global coverage of the surface ocean, they must rely on models to estimate subsurface processes. The proliferation of autonomous assets, such as the Biogeochemical-Argo program (Claustre et al. 2020), complements both ship-based and satellite observations by allowing for vertical resolution of these processes over unprecedented spatiotemporal scales and temporally decoupled systems. Here we leverage a suite of autonomous platforms to demonstrate the observational capabilities across a range of ecosystem states, measuring vertically resolved portions of the BCP such as GPP, NCP, and an estimate of the carbon flux. The organic carbon flux presented here aims to refine the distinction between NCP and export to provide a finer-scale understanding of the BCP, particularly under non-steady-state conditions and over shorter timescales.

The EXport Processes in the Ocean from RemoTe Sensing (EXPORTS) program represents a concerted effort to quantify carbon export to the deep ocean. The central hypothesis is that the fate of primary productivity can be predicted from surface

ecosystem properties, and that the importance of export pathways should vary systematically among ecosystem conditions. The program aimed to gain mechanistic insights into these productivity and export pathways across a range of ecosystem states by choosing two end-member study sites with opposing ecosystem states and BCP strengths: low export efficiency in the late summer Northeast Pacific and high efficiency during the North Atlantic spring bloom (Siegel et al. 2016).

Primary productivity at the low export endmember site (Site 1, North Pacific) is constrained by high stratification and iron limitation (Martin and Fitzwater 1988; Harrison 2002; Meyer et al. 2022), and growth rates are further balanced by grazing (McNair et al. 2021). The low export (< 15%) and high attenuation (~ 70%) of POC result in ~ 5% of net primary productivity (NPP) persisting 100 m below the EZ (Siegel et al. 2016). At the high export site (Site 2, North Atlantic), the alleviation of light limitation catalyzes the spring bloom, further fueled by the entrainment of nutrients by storm activity (Johnson et al. 2024). Nearly half of NPP is exported from the surface, and weak attenuation of POC results in roughly 40% of NPP remaining 100 m below the EZ (Buesseler and Boyd 2009; Briggs et al. 2018; Buesseler et al. 2020b). Consequently, the BCP in this region is an integral part of the global carbon cycle, representing roughly 1.3 Pg C out of the ~ 5–12 Pg C exported annually (Sanders et al. 2014).

Methods

The first campaign occurred in late summer (August 14–September 9, 2018), in the Northeast Pacific near Ocean Station Papa (Site 1, 50°N, 145°W). The second campaign occurred in late spring (May 1–31, 2021) in the northeastern North Atlantic, approximately 170 km east of the Porcupine Abyssal Plain Sustained Observatory (Site 2, 49°N, 16.5°W) (Fig. 1). For clarity, the North Pacific site will hereafter be referred to as Site 1, and the North Atlantic as Site 2. Sampling operations were divided into three separate epochs of 6–9 d to capture the propagation of particles from the surface to 500 m with an assumed sinking speed of 5–10 m d⁻¹. During the North Atlantic campaign, epochs were punctuated by numerous storm events that inhibited ship-based operations. These storms occurred from May 7–11, May 14–16, May 18–21, and May 22–24, and significantly impacted hydrographic and biogeochemical properties. The region experiences strong physical dynamics, with advective processes capable of influencing upper ocean biogeochemical properties on temporal and spatial scales comparable to biological processes (Johnson et al. 2024). Consequently, the campaign was centered in a highly retentive mesoscale anticyclonic eddy to minimize the impact of lateral advection and execute this study in a Lagrangian framework.

Data sources

The North Pacific campaign utilized data from three autonomous gliders and a Biogeochemical-Argo float (World

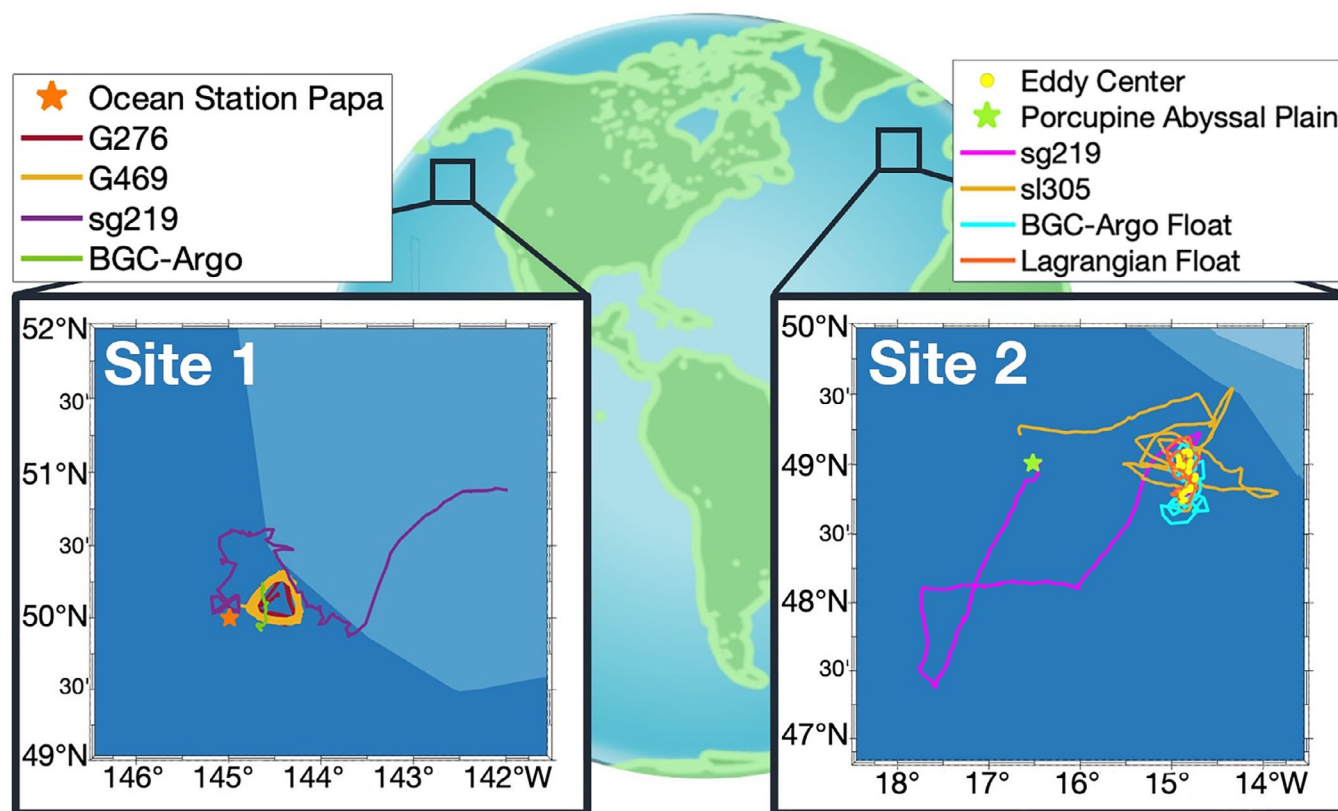


Fig. 1. Location of the two EXport Processes in the Ocean from RemoTe Sensing field campaigns in the North Pacific near Ocean Station Papa (Site 1, August–September 2018) and the North Atlantic, approximately 170 km east of the Porcupine Abyssal Plain (Site 2, May 2021). Positions of autonomous assets before and during the cruise period are also shown.

Meteorological Organization ID: 5905988), retrieved from the Argo Global Data Assembly Center. The North Atlantic campaign combined data from two gliders, one Biogeochemical-Argo float (World Meteorological Organization ID: 1902304), and an instrumented Lagrangian float (D'Asaro 2003) to make a merged dataset of observations in the eddy core, defined as being within 15 km of the eddy center (Erickson et al. 2022). Details on sensor packages and sampling strategy for both campaigns can be found in Supporting Information, Section S.1. Oxygen sensors were calibrated using discrete samples processed by Winkler titration (Dickson 1995). Chlorophyll fluorescence underwent corrections for non-photochemical quenching and calibration using bottle samples. Details concerning sensor calibration and asset intercalibration are available in Supporting Information, Section S.2.

This study utilized publicly available data including 6-h barometric pressure, 10-m wind speed and direction from the National Oceanic and Atmospheric Administration Ocean Station Papa (Site 1) mooring and National Centers for Environmental Prediction reanalysis from the eddy center at Site 2; 6-h wind stress fields from the National Oceanic and Atmospheric Administration Fleet Numerical Meteorology and Oceanography Center; a monthly

climatology of diapycnal diffusivity at Ocean Station Papa (Site 1) from Cronin et al. (2015); summertime diapycnal diffusivity in the North Atlantic (Van Haren et al. 2021); and ship-board measurements collected during the two cruises. Mixed layer depth was calculated using a potential density increase of 0.03 kg m^{-3} relative to the 10-m density (De Boyer Montégut 2004).

Gross and net primary productivity

Gross primary productivity was calculated by fitting a photosynthesis model to the diel cycles of oxygen (O_2) in the ML following Nicholson et al. (2015) and Barone et al. (2019). In this process, the biogenic O_2 signal is isolated from the variation due to physical processes and fits to a model that incorporates responses to light saturation and photo inhibition. The Durbin-Watson test was used to select only days with statistically significant fits ($p < 0.1$). Net primary productivity was measured by onboard incubations using the ^{14}C method, and daily integrated values were calculated to the 1% photosynthetically active radiation depth using a trapezoidal integration (Site 1: $n = 48$; Site 2: $n = 55$, J. Fox personal comms) (Steeman Nielsen 1952; Fox et al. 2020).

Net community production

Vertically resolved NCP was measured down to 100 m using a mass balance model of dissolved O₂ (Emerson 1987; Emerson et al. 2008; Nicholson et al. 2008). This approach separates the net biological flux from physical fluxes and air-sea gas exchange. The net biological flux reflects the balance between productivity and respiration and is equal to NCP. An example of the relative contribution of each of these fluxes to NCP can be found in Supporting Information, Fig. S5. The physical flux was decomposed into components accounting for vertical diffusive, advective, and entrainment fluxes. Horizontal advection of O₂ was assumed to be negligible at Site 1 and minimized at Site 2 by executing this study within a single retentive eddy. Analysis was additionally confined to the period of May 5–June 23, 2021, due to the dissipation of this eddy shortly after. Gas exchange included both diffusive and bubble fluxes. A detailed description of the model and rationale for the parameterizations can be found in Supporting Information, Section S.4.

Uncertainty estimate

A Monte Carlo approach was used to determine the overall error in the NCP term by varying the uncertainties of individual terms (Table 1) in a Gaussian manner over 1000 iterations and taking the standard deviation as the final error estimate. Assigned errors followed literature values and included sensor uncertainty, variability in stoichiometric ratios, and parameterized abiotic processes such as the magnitude of the gas flux components (Cronin et al. 2015; Emerson et al. 2019; Haskell et al. 2020). The O₂-derived productivity terms are converted to carbon units using a modified Redfield ratio for O₂ : C (1.4 : 1; Laws 1991), but this ratio is known to be variable and dependent on factors including the phytoplankton community, respiratory pathways, and nutrient availability (Moreno et al. 2020).

Fate of net community production

The organic matter fixed by NCP in the surface ocean may follow numerous fates: (1) export as POC via gravitational sinking, (2) export via vertical physical transport (e.g., particle injection via subduction), (3) biologically mediated active export such as by diel vertical migration or large mesopelagic

migrators, or (4) accumulation in the surface as DOC or suspended POC, followed by any of the above pathways. The NCP represents the maximum amount of carbon available for export, but over shorter observational timescales, some of this organic carbon follows (4) and accumulates in the water column. The inventory change of POC in the water column can be described as follows:

$$\frac{\Delta \text{POC}}{\Delta t} = \overbrace{(\text{NCP} - f\text{DOC}_{\text{prod}})}^{\text{POC Production}} - \overbrace{(f\text{DVM} + f\text{POC}_{\text{sink}})}^{\text{POC Removal}} - \underbrace{\int_{t_0}^t w_z (\overline{\text{POC}} - \text{POC}_z)}_{\text{Net Advection}} \quad (1)$$

Rearranging to solve for the partitioning of organic matter resulting from NCP as follows:

$$\text{NCP} = \frac{\Delta \text{POC}}{\Delta t} + f\text{DOC}_{\text{prod}} + f\text{DVM} + f\text{POC}_{\text{sink}} + \Delta \text{POC}_{\text{adv}} \quad (2)$$

where NCP is equal to the change in POC inventory ($\frac{\Delta \text{POC}}{\Delta t}$), net DOC production ($f\text{DOC}_{\text{prod}}$), particles moved via diel vertical migration ($f\text{DVM}$), sinking POC flux ($f\text{POC}_{\text{sink}}$), and advective fluxes of POC ($\Delta \text{POC}_{\text{adv}}$). Optical proxies for suspended POC were used to calculate the inventory change and thus improve the estimate of the fate of NCP over shorter timescales. Here we define an organic carbon flux (fC_{org}), comprised of $f\text{DVM}$, $f\text{POC}_{\text{sink}}$, and $f\text{DOC}_{\text{prod}}$, separating out the accumulated suspended POC from sinking fluxes and net DOC production.

$$\begin{aligned} fC_{\text{org}} &= f\text{DOC}_{\text{prod}} + f\text{DVM} + f\text{POC}_{\text{sink}} \\ &= \text{NCP} - \frac{\Delta \text{POC}}{\Delta t} - \Delta \text{POC}_{\text{adv}} \end{aligned} \quad (3)$$

This term, while an imperfect measure of carbon export, leverages autonomous observations to improve our understanding of the partitioning of NCP. These fluxes were evaluated as a mean rate over the entire cruise period. The NCP records the net production or consumption of DOC but does not inform on the vertical transport of DOC. However, an advantage and motivation for this formulation is that $f\text{DOC}_{\text{prod}}$ can be inferred by differencing with independent measurements of the vertical particulate flux. Specifically, we compare to the thorium-234 (^{234}Th)-derived particulate flux (fC_{Th}), which equals the sum of $f\text{DVM}$ and $f\text{POC}_{\text{sink}}$.

$$fC_{\text{Th}} = f\text{DVM} + f\text{POC}_{\text{sink}} \quad (4)$$

Combining Eqs. (3) and (4):

$$f\text{DOC}_{\text{prod}} = fC_{\text{org}} - fC_{\text{Th}} \quad (5)$$

This additional insight into DOC production is valuable because the only direct in situ measurement of DOC is via

Table 1. Uncertainty estimates included in the Monte Carlo simulation to determine overall uncertainty in net community production, taken from the referenced literature.

Source	Error
O ₂ saturation	± 0.5%
Gas exchange model, diffusive velocity (K_S)	± 10%
Gas exchange model, bubble velocities (K_C and K_B)	± 30%
Vertical/Ekman pumping velocity	± 50%
Eddy diffusivity coefficient	± 100%
$\Delta \text{O}_2 : \Delta \text{C}$ ratio	± 0.15

colored dissolved organic matter, which cannot be quantitatively related to bulk DOC (Siegel et al. 2002).

Results

Site 1: North Pacific

Characteristic late summer hydrographic conditions were observed at Site 1 with a shallow ML, strong stratification, weak horizontal hydrographic gradients, abundant macronutrients, and low phytoplankton concentrations. Despite typical concentrations of nitrate and silicate (Fig. 2; Supporting Information, Section S.5), productivity was lower than climatological values. This suggests active iron limitation during the time of station occupation, as surface concentrations never exceeded 0.2 nM (Buck et al. 2023). Low ML chlorophyll *a* (Chl *a*) concentrations allowed light to penetrate farther, deepening the EZ (defined as the 1% photosynthetically active radiation isolume) to 78 ± 6 m ($n = 149$) compared to the historic value of 55 ± 10 m ($n = 10$) (Siegel et al. 2021). Wind speeds remained relatively low and consistent during the cruise period, ranging from 2 to 15 m s^{-1} (Supporting Information, Fig. S3). The ML gradually deepened by ~ 10 m over the course of the cruise, overlying a shallow seasonal thermocline. The Chl *a* and O_2 maxima occurred just below the ML, effectively capped by the warmer surface waters (Fig. 3b, c). Intense stratification continued past the cruise period, as evidenced by the thermal profile and Brunt-Vaisala frequency (Fig. 3a; Supporting Information, Fig. S4).

Vertically integrated Chl *a* demonstrates three small bloom events during the cruise, ranging from 16 to 21 mg m^{-2} . This general bloom pattern is captured in the productivity (NCP and GPP) record (Fig. 4; Supporting Information, Fig. S6), though the O_2 -based NCP and GPP lack the finer temporal scale of the Chl *a* and nutrient data due to an integration period on the order of weeks. The system shifted from net heterotrophic (NCP < 0) to autotrophic (NCP > 0) around the peak of the first bloom in Epoch 1 and remained lightly autotrophic after ship occupation. Seasonal NCP patterns and magnitudes aligned with previous studies, showing peak autotrophy in the summer months (June–August) and a decline to heterotrophy in late fall (Supporting Information, Section S.5, Fig. S2). Cruise mean GPP was roughly half of previous years (Supporting Information, Section S.5, Table S2). Overall, the cruise captured the decline in summertime production and transition towards fall conditions.

Site 2: North Atlantic

Site 2 was highly dynamic and characterized by high productivity, an active eddy field, and frequent storm activity. Low stratification and deep mixing (Fig. 3a) stabilized a couple of weeks before the cruise began. Sampling of the cohesive eddy structure began in late April, at which time net surface warming drove ML shoaling that stabilized the upper ocean and formed a warm surface layer (Supporting Information, Fig. S4). Large storms brought high winds and heavy seas during the cruise period, reaching sustained winds of 20 m s^{-1} .

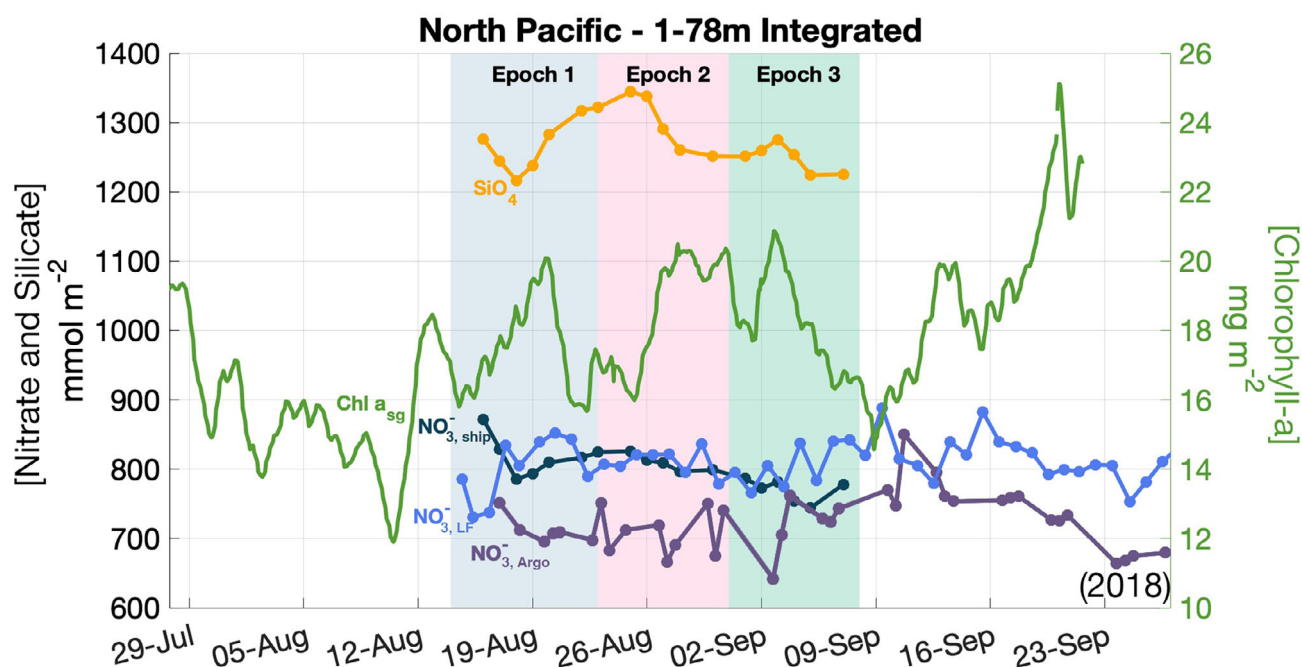


Fig. 2. Vertically integrated euphotic zone properties during the North Pacific cruise (Site 1). On the left axis, silicate (SiO_4) and nitrate from bottle samples ($\text{NO}_3^-_{\text{ship}}$) are shown, as are nitrate data from a nearby Biogeochemical-Argo float ($\text{NO}_3^-_{\text{Argo}}$) and the Lagrangian float ($\text{NO}_3^-_{\text{LF}}$). Corrected chlorophyll *a* (Chl *a*_{sg}) measured by the Seaglider is shown on the right axis.

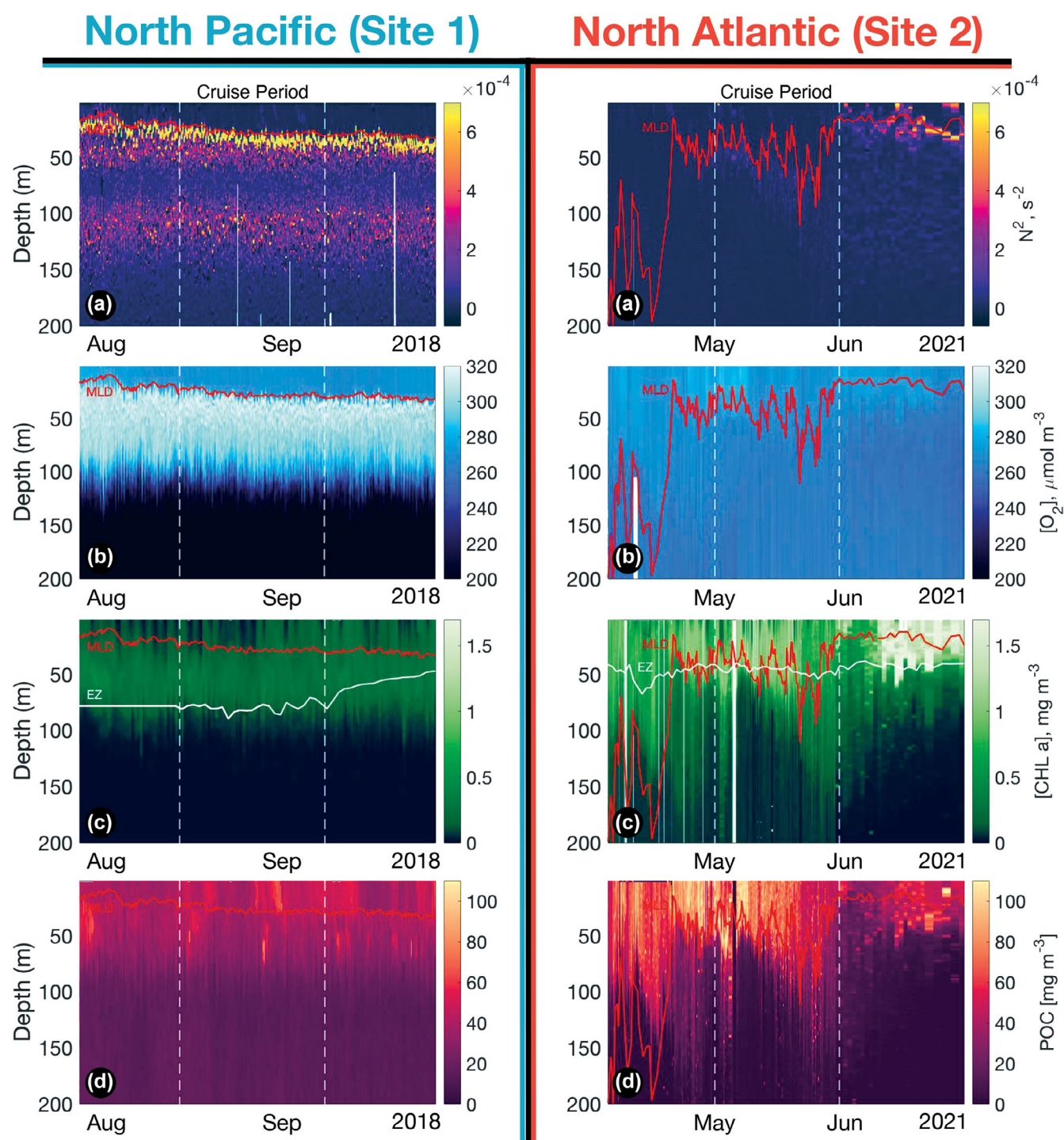


Fig. 3. Hydrographic and biogeochemical properties of the sites before, during, and after occupation, with the North Pacific (Site 1) on the left and North Atlantic (Site 2) on the right. Brunt–Vaisala frequency squared (N^2) is shown in (a), followed by oxygen concentration (b), chlorophyll *a* (c), and particulate organic carbon (POC) (d). The cruise period is demarcated by vertical white dashed lines. The same color scales are utilized between the two sites to highlight the contrast in magnitude. The mixed layer depth (MLD) is overlain in either white, black, or red; the euphotic zone (1% photosynthetically active radiation) depth is plotted in cyan on the chlorophyll sections.

(Supporting Information, Fig. S3). These storm events accounted for 42% of the campaign and periodically deepened the ML down to 56–80 m, which shoaled to ~30 m in the interim

periods. These events homogenized variables and exchanged material with the deeper water column. The mean EZ depth was 48 ± 7 m ($n = 372$) over the cruise period, significantly

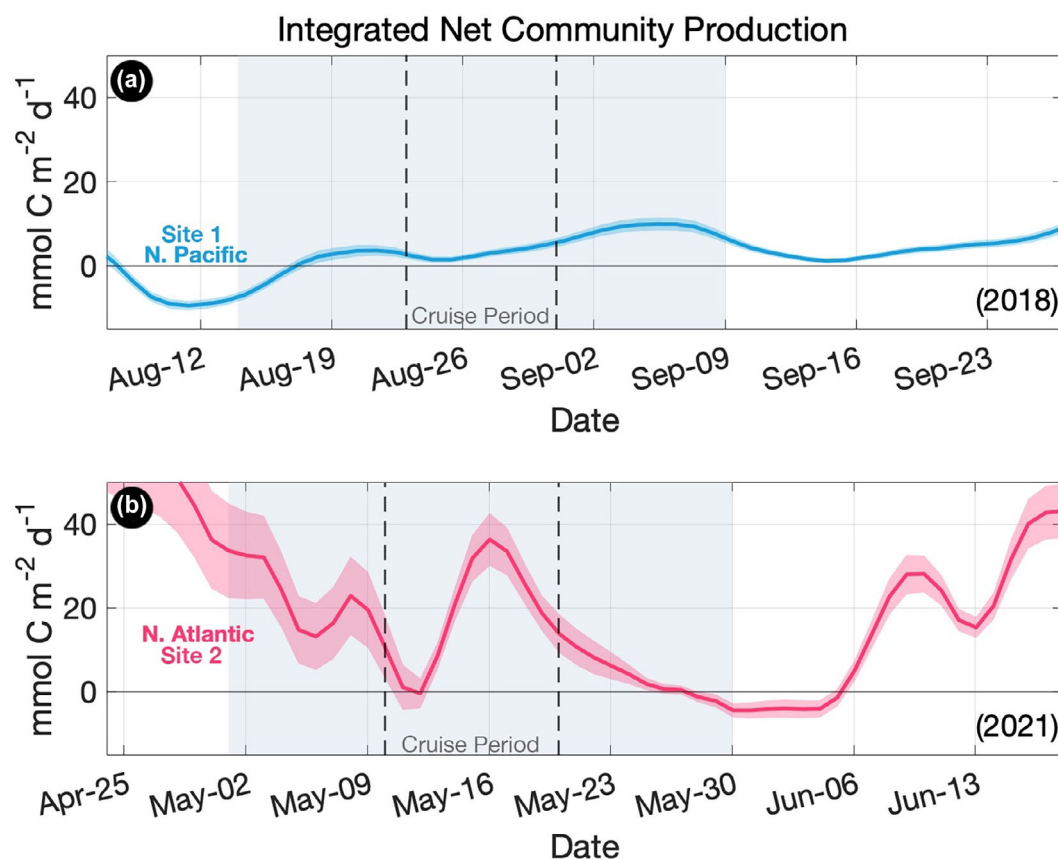


Fig. 4. Cruise period net community production, integrated from the surface to 100 m. The cruise period is depicted in gray shading, and epochs are shown by vertical dashed lines. In (a), the North Pacific site (Site 1) is shown in blue; in (b) the North Atlantic site (Site 2) is shown in pink. Shading indicates uncertainty bounds derived from the Monte Carlo simulation and is greater at Site 2 due to the greater contribution of bubble fluxes.

shallower than at Site 1. O_2 concentrations in the surface ocean were strongly influenced by the high wind-driven gas exchange and reflected deep mixing from previous seasonal pycnoclines (Fig. 3b). Chlorophyll *a* concentrations were generally an order of magnitude higher than at Site 1 and constrained to the ML, though storm events similarly mixed this signal downward (Fig. 3c).

Euphotic zone Chl *a* captured a sustained peak in productivity during the middle of the cruise period, beginning in the middle of Epoch 1 and declining by the end of Epoch 2 (Fig. 5). Values peaked around 75 mg m^{-2} , a slightly lower magnitude than a subsequent bloom after the end of site occupation. Levels of biogenic silica and nitrate fell after the Chl *a* peak in the first few days of May, while silicate steadily increased. The sustained productivity signal seen in the Chl *a* record is reflected in GPP, NPP, and NCP (Fig. 8). Gross primary productivity increased from Epoch 1 to Epoch 2 and remained high into Epoch 3, while NPP remained stable before a decrease in the third epoch. The time series of NCP (Fig. 4) shows an increase alongside the increasing Chl *a* concentrations, and a similar decline as the bloom weakened in the 2nd and 3rd epochs, though the system remained net autotrophic. The vertical profile of volumetric NCP shows high daily variability,

even reaching orders of magnitude of difference during the month-long cruise (Supporting Information, Fig. S7). While the ML was generally lightly autotrophic and peaks in productivity occasionally occurred near the base of the ML, no consistent structure was seen.

Carbon export fluxes

Export was estimated via a defined organic carbon flux ($f_{\text{C}_{\text{org}}}$; Eq. 3), which removes changes in the POC inventory ($\Delta\text{POC}_{\text{inv}}$), corrected for advective fluxes, from the cumulative NCP (Fig. 6). During the field campaigns, three times as much fixed carbon (as cumulative NCP) was produced at Site 2 than at Site 1, and the impact of $\Delta\text{POC}_{\text{inv}}$ differed greatly. At Site 1, $\Delta\text{POC}_{\text{inv}}$ was positive, increasing, and of similar magnitude as NCP, demonstrating that most of the carbon remained in the upper water column with little detectable export. At Site 2, cumulative NCP vastly outweighed $\Delta\text{POC}_{\text{inv}}$, which ranged from -310 to 90 mmol C m^{-2} . By mid-May, inventory began decreasing, which may be attributable to enhanced sinking export of both new and old POC, increased dissolution to DOC, or poorly constrained lateral advection. Mean $f_{\text{C}_{\text{org}}}$ rates at Site 2 ($20.3 \pm 2.3 \text{ mmol C m}^{-2} \text{ d}^{-1}$) were six times higher than at Site 1 ($3.4 \pm 0.7 \text{ mmol C m}^{-2} \text{ d}^{-1}$) when evaluated at 100 m (Fig. 7). In

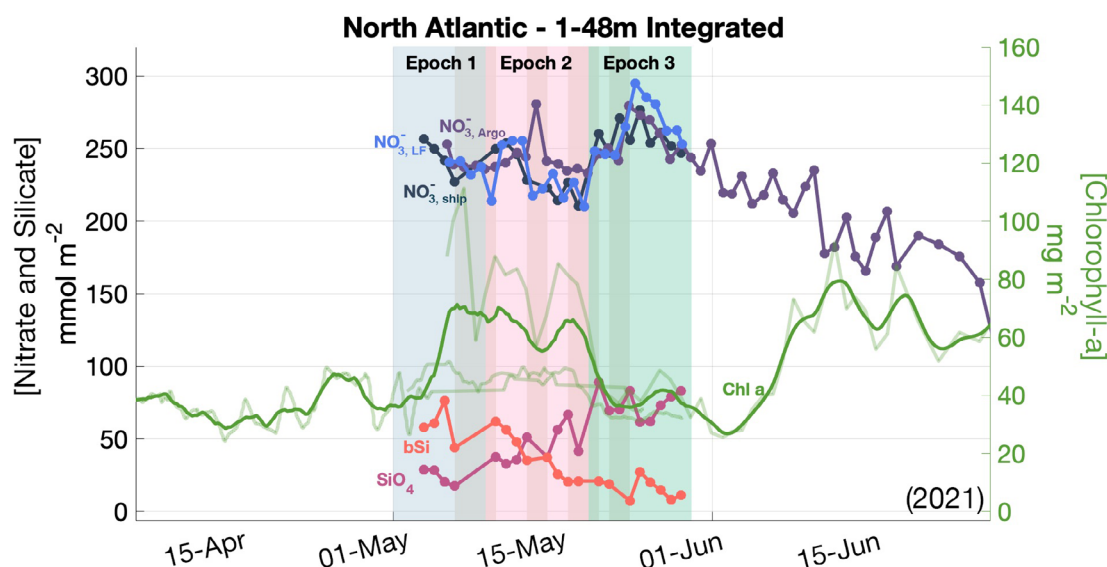


Fig. 5. Vertically integrated euphotic zone properties during the North Atlantic campaign. Epochs are separated by colored bars, and additional shaded bars correspond to storm events during the cruise. Corrected chlorophyll *a* (Chl *a*) from two gliders and a Biogeochemical-Argo float is plotted on the right axis, with the 12-pt moving mean value in a darker green line. On the left axis, nitrate inventories from a Biogeochemical-Argo float ($\text{NO}_3^-_{\text{Argo}}$), the Lagrangian float ($\text{NO}_3^-_{\text{LF}}$), and bottle samples ($\text{NO}_3^-_{\text{ship}}$) tracked closely. Silicate inventories (SiO_4) and biogenic silica (bSi) were inversely correlated.

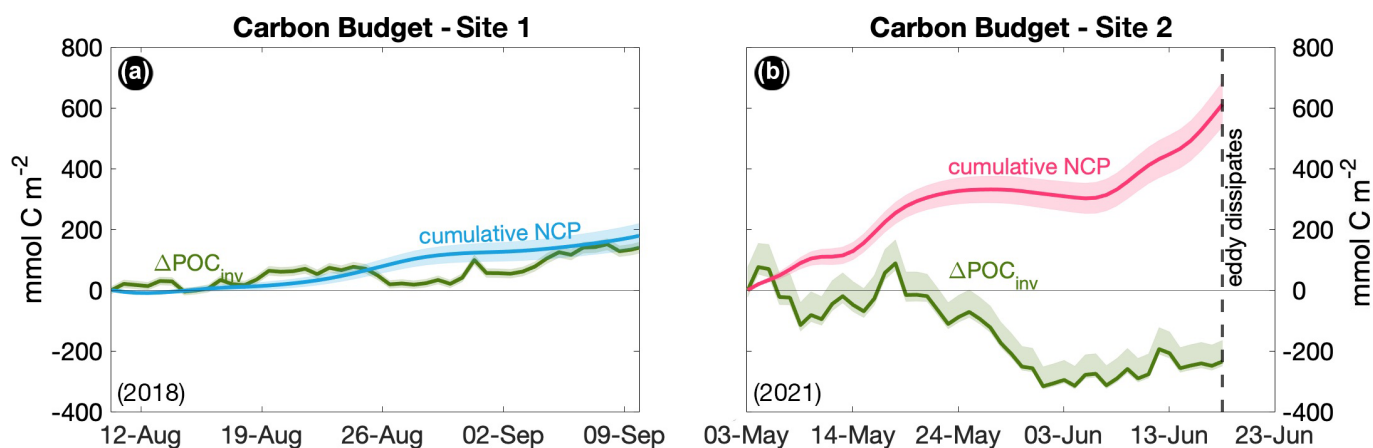


Fig. 6. Cumulative cruise period net community production and the accompanying advection-corrected changes in the particulate organic carbon (POC) inventory ($\Delta\text{POC}_{\text{inv}}$), integrated from 0 to 100 m in (a) the North Pacific (Site 1) and (b) the North Atlantic (Site 2). Shading depicts the uncertainty range as calculated by the Monte Carlo simulation.

the low export setting (Site 1), fC_{org} and fC_{Th} were equal at both depths, while at Site 2, fC_{org} was always greater. The difference between these two fluxes is attributed to the net production of DOC ($f\text{DOC}_{\text{prod}}$) (Eq. 5). At Site 1, this was effectively zero at both depths, but at Site 2 ranged from $18.6 \pm 10.0 \text{ mmol C m}^{-2} \text{ d}^{-1}$ at 60 m and $8.7 \pm 3.5 \text{ mmol C m}^{-2} \text{ d}^{-1}$ at 100 m.

The BCP efficiency has commonly been measured in terms of the *ef*-ratio ($\text{NCP} : \text{NPP}$) (Laws et al. 2000) and the *ez*-ratio ($f\text{POC}_{\text{sink}} : \text{NPP}$) (Buesseler and Boyd 2009). These ratios were calculated using the autonomously measured NCP and fC_{org} (Table 2) and both reflect higher export efficiency at Site 2.

Discussion

Contrasting conditions between sites

Two contrasting ecosystem states were successfully captured with the field campaigns. They occurred during different seasons and years and are not intended to be broadly representative of their respective regions. Instead, they represent distinct endmember states that occur across the ocean at various times and locations, offering insight into the interplay of biogeochemical and physical forcing for broader applications. Physical forcing impacted the distribution of several biogeochemical variables, such as dissolved O_2 , nitrate, and POC

concentration. At Site 1, strong stratification restricted productivity in the ML, which peaked immediately below its base as evidenced by maxima of O₂, Chl *a*, and volumetric NCP (Supporting Information, Fig. S7). When averaged by epoch, neither GPP nor NPP significantly changed over the cruise period despite a ninefold increase in NCP (Fig. 8). This increase in NCP corresponds with increasing Chl *a* concentrations in the middle of Epoch 2, which may have arisen from the gradual deepening of the ML entraining more iron (increasing from 0.04 to 0.20 nM) and a subsequent increase in new production. In

contrast to the stable structure at Site 1, the surface density gradient at Site 2 was relatively small and primarily driven by the thermal gradient. The salinity profile had small vertical gradients and an inverted structure, placing saltier water over fresher, largely due to horizontal advection (Johnson et al. 2024) (Supporting Information, Fig. S4). The site was consequently well-ventilated and had smaller vertical gradients in O₂ and Chl *a* (Fig. 3b, c). Surface nitrate at Site 1 slowly increased with the deepening ML, while deep mixing during storms drove concentrations at Site 2. The impact of physical mixing was underlined by the dwindling of post-cruise nitrate inventories at Site 2 due to biological drawdown as meteorological conditions calmed. The EZ-integrated Chl *a* was three to four times higher at Site 2 than at Site 1, and nitrate concentrations were 2–3 times lower. Silicate concentrations at Site 1 were 14 times higher than at Site 2, which was largely silicate limited. This further underscores the fact that despite ample macronutrients, productivity at Site 1 was limited by iron. Site 1 had greater

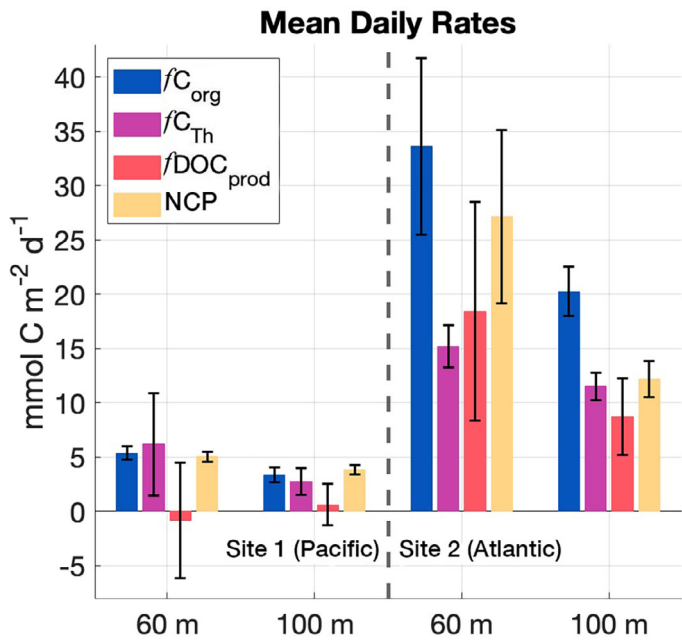


Fig. 7. Comparison of mean daily organic carbon flux (*f*C_{org}, blue bars) as calculated by this study vs. the thorium-234 derived POC flux (*f*C_{Th}, purple bars) at 60 m and 100 m at both study sites. Error bars depict the 95% confidence interval. The left-hand side depicts measurements from the North Pacific site (Site 1), and the right-hand side is the North Atlantic site (Site 2). The inferred net dissolved organic carbon (DOC) production (*f*DOC_{prod}, orange bars) is calculated as the difference between the two measurements (*f*C_{org} – *f*C_{Th}). At times, net community production (yellow bars) is smaller than the export terms, which suggests a contribution to export from earlier productivity that is consistent with the cruise period beginning on the tail end of a previous bloom.

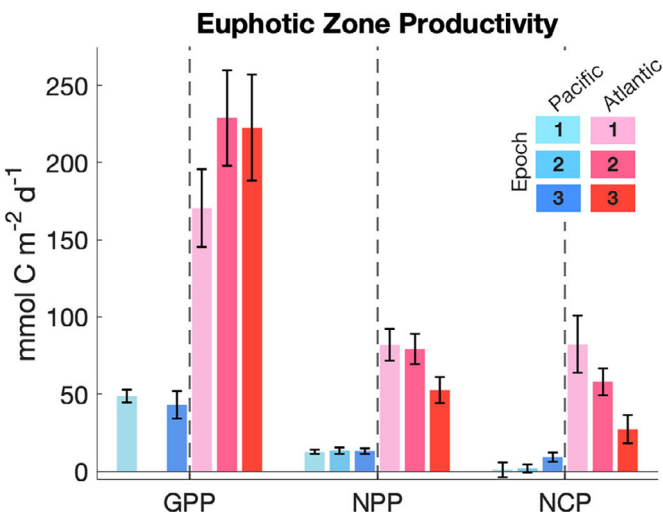


Fig. 8. Daily rate of gross primary production (GPP), net primary production (NPP), and net community production (NCP) at each field site (Site 1, blue; Site 2, pink). Error bars represent the 95% confidence interval. From left to right, each bar represents the epoch average. Studies occurred during different seasons and years and capture a different endmember ecosystem state at each respective site.

Table 2. Cruise-averaged productivity terms and export ratios evaluated at the base of the euphotic zone, summarized for both study sites. The *ef*-ratio is net community production (NCP) : net primary productivity (NPP) and the *ez*-ratio in this study was calculated using *f*C_{org} : NPP rather than *f*POC_{sink}.

Term	Site 1	Site 2
Gross primary productivity	44.2 ± 24.7 (mmol C m ⁻² d ⁻¹)	210.4 ± 19.3 (mmol C m ⁻² d ⁻¹)
NPP	13.0 ± 1.9 (mmol C m ⁻² d ⁻¹)	69.1 ± 16.5 (mmol C m ⁻² d ⁻¹)
NCP	4.1 ± 5.2 (mmol C m ⁻² d ⁻¹)	52.8 ± 24.5 (mmol C m ⁻² d ⁻¹)
<i>ef</i> -ratio at 1% photosynthetically active radiation (PAR)	0.33 (78 m)	0.73 (48 m)
<i>ez</i> -ratio at 1% PAR	0.24 (78 m)	0.69 (48 m)

background concentrations of POC, particularly concentrated in the EZ, while Site 2 had higher concentrations in the surface ocean but fewer suspended particles in the mesopelagic (Fig. 3d). Overall, Site 1 displayed calm weather and high stratification that prevented nutrients from mixing into the ML, hampering productivity. By contrast, the study near Site 2 was punctuated by intense storms that initially disrupted the bloom but upwelled nutrients and stimulated primary productivity.

Differing bloom dynamics prevailed at the two sites. Site 1 demonstrated a cyclic pattern where the termination of one bloom fertilizes the EZ with micronutrients and stimulates a subsequent bloom (Fig. 2). Site 2 aligned instead with the second phase of a dual-phase bloom observed in other studies in the region, such as the JGOFS North Atlantic Bloom Experiment and 2008 North Atlantic Bloom Experiment (Sieracki et al. 1993; Alkire et al. 2014; Cetinić et al. 2015). The start of the cruise reflected the decline of a large diatom bloom with decreasing NCP (Fig. 4) and the biogenic silica inventory decreased as it remineralized to silicate. This first phase depleted silicate concentrations but left ample nitrate, which in turn supported a second bloom of non-silicified phytoplankton (Cohn et al. 2023; Meyer et al. 2023). This phase peaked midway through the cruise, reflected by peaks in Chl *a*, GPP, and NCP (Supporting Information, Fig. S6). As the bloom waned, nitrate stocks were replenished, though additionally due to deep mixing and storm-driven entrainment of nutrients. This then fueled elevated Chl *a* and NCP in early June, after the end of site occupation. Deep mixing infusing nutrient-rich waters into the EZ appears to be a primary driver of the higher export ecosystem, as it enables new production and facilitates its removal from the surface ocean.

While both campaigns showed variability during the cruise period, overall productivity differed significantly between the two sites. At Site 1, cruise-averaged GPP and NPP were roughly a fifth of that observed at Site 2 (Table 2). Consequently, NPP was roughly 30% of GPP at both sites. Notably, the mean daily NCP at Site 2 was 13 times greater than at Site 1 and a larger fraction of NPP (*ef*-ratio). The vertical structure of NCP at Site 1 was stable, peaking directly below the heterotrophic ML and ranging from -2 to 2 mmol C m⁻³ d⁻¹ (Supporting Information, Fig. S7). A corresponding peak in POC flux was observed at 50 m, 15 m below the productivity peak (Buesseler et al. 2020a). Site 2 lacked consistent vertical structure, though autotrophy was most pronounced above 50 m and NCP varied from -20 to 16 mmol C m⁻³ d⁻¹. The time series of NCP (Fig. 4) revealed a shift from heterotrophy to light autotrophy at Site 1 a few days into the cruise, with NCP gradually building, whereas Site 2 showed large peaks of NCP crashing into near heterotrophy as the next bloom began to build. In general, Site 2 exhibited significantly higher primary production than Site 1 and experienced fewer heterotrophic controls, allowing more carbon to remain fixed and available for export.

Linking NCP to export

The multi-tracer approach applied in this study highlights the importance of multiple fates of NCP to both particulate and dissolved pools, providing an upper bound on exportable carbon. Furthermore, it illuminates the limitation of export ratios calculated only from the ratio of sinking POC to NPP (*ez*-ratio). The *ez*-ratios reported in this study (Table 2) exceed literature values of approximately 0.15 for Site 1 and 0.45 for Site 2 (Buesseler and Boyd 2009; Siegel et al. 2016), which are derived using $f_{\text{POC}_{\text{sink}}}$ measurements from ²³⁴Th or sediment traps. During bloom conditions, NPP temporally leads particulate export, so it is not surprising that *ez*-ratio snapshots collected during blooms will underestimate long-term averages. The *ez*-ratios are blind to seasonal accumulation of POC and DOC, while, in contrast, our approach provides a more direct assessment of the flow of carbon through upper ocean ecosystems, tracking a larger pool of carbon available for export. Our results are more consistent with early results based on new production (Eppley and Peterson 1979) which shows a positive correlation between new production and NPP.

A steady-state assumption is inadequate for many ocean regions and observational timescales. For instance, at Site 2 (North Atlantic), the cohesive eddy structure persisted only 3 weeks beyond the cruise period. Removing increases in POC stock from NCP allows for an improved estimate of the export during the month-long cruise period. Comparing $f_{\text{C}_{\text{org}}}$ to the traditional ²³⁴Th method, which only captures removal from the particulate pool (Eq. 4), illuminated the important role of DOC production at the high export site (Site 2). However, this method cannot differentiate between net DOC production, accumulation, and export, which depend on the vertical and lateral movement of water masses, nor can it offer insight into the production mechanism.

Small export fluxes were observed at Site 1, and $f_{\text{C}_{\text{org}}}$ and $f_{\text{C}_{\text{Th}}}$ were approximately equal at both depths. This suggests that organic matter was primarily exported through gravitational sinking and DVM with no significant $f_{\text{DOC}_{\text{prod}}}$ detected, in agreement with findings by Huang et al. (2022). Cumulative cruise period $f_{\text{DOC}_{\text{prod}}}$ (0–100 m) amounted to 17.6 ± 48.3 mmol C m⁻², or a meager increase of 0.2 ± 0.5 μM if all DOC accumulated. At Site 2, $f_{\text{C}_{\text{org}}}$ was roughly twice $f_{\text{C}_{\text{Th}}}$, implying an enhanced contribution of $f_{\text{DOC}_{\text{prod}}}$, particularly at the shallower integration depth where $f_{\text{C}_{\text{Th}}}$ was only 50–60% of NCP. By 100 m, NCP and $f_{\text{C}_{\text{Th}}}$ were in balance, suggesting that nearly all of NCP in the upper 100 m was exported as POC over the month-long observational period. The export of both new and existing POC is captured by both $f_{\text{C}_{\text{Th}}}$ and $f_{\text{C}_{\text{org}}}$, with $f_{\text{C}_{\text{org}}}$ additionally containing $f_{\text{DOC}_{\text{prod}}}$. The NCP reflects only new POC and $f_{\text{DOC}_{\text{prod}}}$. The export of POC that was produced before NCP measurements began will thus result in $f_{\text{C}_{\text{org}}}$ outweighing NCP, as was seen at Site 2.

Vertical export of DOC is a significant part of the BCP in the North Atlantic, accounting for roughly 0.081 Pg C leaving the upper 100 m and fueling the mesopelagic (Carlson

et al. 2010). Previous work has shown as much as 35% of seasonal NCP accumulating as DOC, with annual estimates of $0.34\text{--}1.15 \text{ mol C m}^{-2}$ of DOC export to the mesopelagic (Baetge et al. 2020), underscoring the importance of considering DOC when assessing the carbon budget in the surface ocean. At Site 2, cumulative production amounted to $243.7 \pm 89.6 \text{ mmol C m}^{-2}$, or an increase in concentration of $2.4 \pm 0.9 \mu\text{M}$ DOC if accumulated across the upper 100 m. Historical estimates of North Atlantic “surplus” DOC accumulation from winter to spring range from 10 to $290 \text{ mmol C m}^{-2}$ (Baetge et al. 2020) between the time of deep convection to late spring, suggesting that our study may have captured a significant portion of net production since deep winter convection.

While DOC production influences the divergence between fC_{org} and fC_{Th} , other factors introduce uncertainty into each measurement. The method presented here relies on despiked backscatter at 700 nm ($b_{\text{bp}700}$), which is most sensitive to particles between 0.5 and $30 \mu\text{m}$ (Dall’Omo et al. 2009; Organelli et al. 2018). Different sinking particles exhibit varying residence times and behaviors, with the proxy’s accuracy influenced by biological processes, mixing, and particle dynamics. The ^{234}Th method captures the ^{234}Th deficit across any particle size class but uses a POC:Th ratio from the midsize class (Site 1: $5\text{--}51 \mu\text{m}$; Site 2: $>5 \mu\text{m}$) as collected by McLane pumps. Variability in this ratio between size classes can impact agreement between methods. Furthermore, while the temporal period NCP integrates is on a similar timescale to the ^{234}Th method (days to weeks), the two are not necessarily coupled. Additionally, backscatter-derived POC inventory change represents daily fluctuations and captures shorter-term or episodic variability in particle fluxes.

Conclusions

The EXPORTS field campaigns effectively captured contrasting endmembers of the BCP. Autonomous assets provided essential contextual information, enabling the characterization of productivity regimes and export efficiency at both sites. The iron-limited Site 1 exhibited 8–36% (mean: 20%) of the productivity observed at Site 2, reduced export, and greater recycling of organic matter in the surface ocean, and a diminished role of DOC production. Less than 10% of GPP remained fixed as NCP available for export, whereas at Site 2, NCP accounted for roughly a quarter of GPP. Export efficiency metrics were two to three times higher at Site 2, highlighting its characterization as a highly productive and efficient regime. Sinking POC fluxes (fC_{Th}) balanced NCP at both 60 and 100 m at Site 1, but only at 100 m at Site 2. By combining our fC_{org} term with ^{234}Th -derived POC fluxes, we inferred DOC production at each site and found a greater contribution at Site 2 than at Site 1. This was particularly true at a shallower depth horizon (60 m) for which 54% of fC_{org} was routed to net DOC production rather than particulate export.

This study demonstrates that sensor-based observations can effectively track the metabolic use of carbon from GPP down to export, capturing the same large-scale patterns as shipboard methods. While most models predict a decline in NPP, the challenges of both observing and modeling the BCP lead to over a 40% uncertainty in the predicted response of the global export flux (Henson et al. 2022). Remote sensing-based parameterizations of export flux and its global distribution are often used as a boundary condition in climate models (Bushinsky and Emerson 2015). Therefore, expanded in situ observation and improved best practices will yield more accurate models and informed policy decisions. The increased spatial and temporal sampling by autonomous platforms, combined with a diffusive tracer like O_2 , can address the patchiness issues often seen with ship-based and particle-based approaches. Additionally, employing a sensor-based POC inventory change correction to NCP estimates can refine carbon export estimates. This was particularly important for measurements at Ocean Station Papa, where POC inventory changes were a similar magnitude to NCP. By integrating these autonomous estimates with ship-based measurements, we gained broader comprehensive insight into the relative contribution of DOC vs. POC production in the ecosystem. At this scale, the change in DOC concentration was too small to be observed using traditional bottle samples, and only through this method can it be observed. Autonomous observations enhance our ability to observe the BCP and, when integrated with other observational strategies, deepen our understanding of the ocean’s role in the global carbon cycle.

Author Contributions

Shawnee Traylor participated in the field deployments, analyzed the data, and prepared the manuscript. David P. Nicholson participated in the conceptualization, methodology, resources, and writing—review and editing. Samantha J. Cleverger contributed to data collection, data processing, and writing—review and editing. Ken O. Buesseler contributed to data collection, data processing, and writing—review and editing. Eric D’Asaro participated in the conceptualization, data curation, methodology, software, and resources. Craig M. Lee participated in the conceptualization, data curation, and resources.

Acknowledgments

The authors acknowledge support from the National Aeronautics and Space Administration Ocean Biology and Biogeochemistry, National Science Foundation Biological and Chemical Oceanography programs, and National Oceanic and Atmospheric Administration Global Ocean Monitoring and Observing program. This work is part of a collaborative effort of the entire EXPORTS team, led by D. Siegel and I. Cetinić. Gratitude is extended to the captains, crews, and ship operations staff of the R/Vs *Revelle* and *Ride*, *RSS Discovery*, and *James Cook*. We are grateful to the EXPORTS Hydro

team for data collection and processing, K. Buck and S. Burns for dissolved iron data, A. Niebergall and W. Tang for sea assistance, and J. Fox, L. Johnson, and Y. Huang for input data analysis. Support from the National Aeronautics and Space Administration was provided as part of the EXPORTS. In particular, Craig M. Lee, Eric D'Asaro, David P. Nicholson, and Shawnee Traylor were supported by the National Aeronautics and Space Administration Grant 80NSSC17K0663. Ken O. Buesseler and Samantha J. Clevenger were supported by the National Aeronautics and Space Administration Grants 80NSSC17K0555 and 80NSSC17K0662. David P. Nicholson and Shawnee Traylor also received support from the National Oceanic and Atmospheric Administration Global Ocean Monitoring and Observing Grant NA19OAR4320074. Additional support came from National Science Foundation Global Ocean Biogeochemistry Array Project under award 1946578, with operational support from award 2110258. Shawnee Traylor is supported by National Science Foundation Graduate Research Fellowship 1745302. Open Access funding enabled and organized by MIT Hybrid 2025.

Conflicts of Interest

None declared.

References

- Alkire, M. B., C. Lee, E. D'Asaro, et al. 2014. "Net Community Production and Export from Seaglider Measurements in the North Atlantic after the Spring Bloom." *Journal of Geophysical Research, Oceans* 119: 6121–6139. <https://doi.org/10.1002/2014JC010105>.
- Baetge, N., J. R. Graff, M. J. Behrenfeld, and C. A. Carlson. 2020. "Net Community Production, Dissolved Organic Carbon Accumulation, and Vertical Export in the Western North Atlantic." *Frontiers in Marine Science* 7: 227. <https://doi.org/10.3389/fmars.2020.00227>.
- Barone, B., D. Nicholson, S. Ferrón, E. Firing, and D. Karl. 2019. "The Estimation of Gross Oxygen Production and Community Respiration from Autonomous Time-Series Measurements in the Oligotrophic Ocean." *Limnology and Oceanography: Methods* 17: 650–664. <https://doi.org/10.1002/lom3.10340>.
- Boyd, P. W., H. Claustre, M. Levy, D. A. Siegel, and T. Weber. 2019. "Multi-Faceted Particle Pumps Drive Carbon Sequestration in the Ocean." *Nature* 568: 327–335. <https://doi.org/10.1038/s41586-019-1098-2>.
- Briggs, N., K. Guðmundsson, I. Cetinić, et al. 2018. "A Multi-Method Autonomous Assessment of Primary Productivity and Export Efficiency in the Springtime North Atlantic." *Biogeosciences* 15: 4515–4532. <https://doi.org/10.5194/bg-15-4515-2018>.
- Buck, K., S. M. Burns, B. D. Jenkins, and M. A. Brzezinski. 2023. "Dissolved Trace Metal (Mn, Fe, Co, Ni, Cu, Zn, Cd, Pb) Concentrations from Surface (Towfish) Samples Collected during the 2018 EXPORTS Cruise (RR1813) to the Subarctic North Pacific near Ocean Station PAPA (Station P) on R/V Roger Revelle." <https://doi.org/10.26008/1912/BCO-DMO.869683.2>.
- Buesseler, K. O., C. R. Benitez-Nelson, M. Roca-Martí, et al. 2020a. "High-Resolution Spatial and Temporal Measurements of Particulate Organic Carbon Flux Using Thorium-234 in the Northeast Pacific Ocean during the EXport Processes in the Ocean from RemoTe Sensing Field Campaign." *Elementa: Science of the Anthropocene* 8: 030. <https://doi.org/10.1525/elementa.2020.030>.
- Buesseler, K. O., and P. W. Boyd. 2009. "Shedding Light on Processes that Control Particle Export and Flux Attenuation in the Twilight Zone of the Open Ocean." *Limnology and Oceanography* 54: 1210–1232. <https://doi.org/10.4319/lo.2009.54.4.1210>.
- Buesseler, K. O., P. W. Boyd, E. E. Black, and D. A. Siegel. 2020b. "Metrics that Matter for Assessing the Ocean Biological Carbon Pump." *Proceedings of the National Academy of Sciences* 117: 9679–9687. <https://doi.org/10.1073/pnas.1918114117>.
- Burd, A. B., D. A. Hansell, D. K. Steinberg, et al. 2010. "Assessing the Apparent Imbalance between Geochemical and Biochemical Indicators of Meso- and Bathypelagic Biological Activity: What the @\$#! Is Wrong with Present Calculations of Carbon Budgets?" *Deep-Sea Research Part II: Topical Studies in Oceanography* 57: 1557–1571. <https://doi.org/10.1016/j.dsr2.2010.02.022>.
- Bushinsky, S. M., and S. Emerson. 2015. "Marine Biological Production from In Situ Oxygen Measurements on a Profiling Float in the Subarctic Pacific Ocean." *Global Biogeochemical Cycles* 29: 2050–2060. <https://doi.org/10.1002/2015GB005251>.
- Carlson, C. A., D. A. Hansell, N. B. Nelson, et al. 2010. "Dissolved Organic Carbon Export and Subsequent Remineralization in the Mesopelagic and Bathypelagic Realms of the North Atlantic Basin." *Deep-Sea Research Part II: Topical Studies in Oceanography* 57: 1433–1445. <https://doi.org/10.1016/j.dsr2.2010.02.013>.
- Cetinić, I., M. J. Perry, E. D'Asaro, et al. 2015. "A Simple Optical Index Shows Spatial and Temporal Heterogeneity in Phytoplankton Community Composition during the 2008 North Atlantic Bloom Experiment." *Biogeosciences* 12: 2179–2194. <https://doi.org/10.5194/bg-12-2179-2015>.
- Ciais, P., C. Sabine, G. Bala, et al. 2013. "Carbon and Other Biogeochemical Cycles." In *Climate Change 2013—The Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, 465–570. Cambridge, UK: Cambridge University Press.
- Claustre, H., K. S. Johnson, and Y. Takeshita. 2020. "Observing the Global Ocean with Biogeochemical-Argo." *Annual Review of Marine Science* 12: 23–48. <https://doi.org/10.1146/annurev-marine-010419-010956>.

- Cohn, M. R., B. Stephens, M. G. Meyer, et al. 2023. "Microbial Respiration in Contrasting Ocean Provinces via High-Frequency Optical Assays." <https://doi.org/10.1101/2023.07.20.549894>.
- Cronin, M. F., N. A. Pelland, S. R. Emerson, and W. R. Crawford. 2015. "Estimating Diffusivity from the Mixed Layer Heat and Salt Balances in the North Pacific." *Journal of Geophysical Research, Oceans* 120: 7346–7362. <https://doi.org/10.1002/2015JC011010>.
- Dall'Olmo, G., T. K. Westberry, M. J. Behrenfeld, E. Boss, and W. H. Slade. 2009. "Significant Contribution of Large Particles to Optical Backscattering in the Open Ocean." *Biogeosciences* 6: 947–967. <https://doi.org/10.5194/bg-6-947-2009>.
- D'Asaro, E. A. 2003. "Performance of Autonomous Lagrangian Floats." *Journal of Atmospheric and Oceanic Technology* 20: 896–911. [https://doi.org/10.1175/1520-0426\(2003\)020<0896:POALF>2.0.CO;2](https://doi.org/10.1175/1520-0426(2003)020<0896:POALF>2.0.CO;2).
- De Boyer Montégut, C. 2004. "Mixed Layer Depth over the Global Ocean: An Examination of Profile Data and a Profile-Based Climatology." *Journal of Geophysical Research* 109: C12003. <https://doi.org/10.1029/2004JC002378>.
- DeVries, T., and T. Weber. 2017. "The Export and Fate of Organic Matter in the Ocean: New Constraints from Combining Satellite and Oceanographic Tracer Observations." *Global Biogeochemical Cycles* 31: 535–555. <https://doi.org/10.1002/2016GB005551>.
- Dickson, A. D. 1995. Determination of Dissolved Oxygen in Seawater by Winkler Titration. Woods Hole, MA, USA: WOCE Operations Manual, Part 3.13 Operations & Methods, WHP Office Report WHPO 91-1.
- Ducklow, H. W., and S. C. Doney. 2013. "What Is the Metabolic State of the Oligotrophic Ocean? A Debate." *Annual Review of Marine Science* 5: 525–533. <https://doi.org/10.1146/annurev-marine-121211-172331>.
- Emerson, S. 1987. "Seasonal Oxygen Cycles and Biological New Production in Surface Waters of the Subarctic Pacific Ocean." *Journal of Geophysical Research* 92: 6535–6544. <https://doi.org/10.1029/JC092iC06p06535>.
- Emerson, S., C. Stump, and D. Nicholson. 2008. "Net Biological Oxygen Production in the Ocean: Remote In Situ Measurements of O₂ and N₂ in Surface Waters." *Global Biogeochemical Cycles* 22: GB3023. <https://doi.org/10.1029/2007GB003095>.
- Emerson, S., B. Yang, M. White, and M. Cronin. 2019. "Air-Sea Gas Transfer: Determining Bubble Fluxes with In Situ N₂ Observations." *Journal of Geophysical Research, Oceans* 124: 2716–2727. <https://doi.org/10.1029/2018JC014786>.
- Eppley, R. W., and B. J. Peterson. 1979. "Particulate Organic Matter Flux and Planktonic New Production in the Deep Ocean." *Nature* 282: 677–680. <https://doi.org/10.1038/282677a0>.
- Erickson, Z. K., E. Fields, M. M. Omand, et al. 2022. EXPORTS North Atlantic Eddy Tracking. Woods Hole, MA: NASA STI Program and Woods Hole Oceanographic Institution.
- Fox, J., M. J. Behrenfeld, N. Haëntjens, et al. 2020. "Phytoplankton Growth and Productivity in the Western North Atlantic: Observations of Regional Variability from the NAAMES Field Campaigns." *Frontiers in Marine Science* 7: 24. <https://doi.org/10.3389/fmars.2020.00024>.
- Hansell, D. A., and C. A. Carlson. 1998. "Deep-Ocean Gradients in the Concentration of Dissolved Organic Carbon." *Nature (London)* 395: 263–266. <https://doi.org/10.1038/26200>.
- Harrison, P. J. 2002. "Station Papa Time Series: Insights into Ecosystem Dynamics." *Journal of Oceanography* 58: 259–264. <https://doi.org/10.1023/A:1015857624562>.
- Haskell, W. Z., A. J. Fassbender, J. S. Long, and J. N. Plant. 2020. "Annual Net Community Production of Particulate and Dissolved Organic Carbon from a Decade of Biogeochemical Profiling Float Observations in the Northeast Pacific." *Global Biogeochemical Cycles* 34: e2020GB006599. <https://doi.org/10.1029/2020GB006599>.
- Henson, S. A., C. Laufkötter, S. Leung, S. L. C. Giering, H. I. Palevsky, and E. L. Cavan. 2022. "Uncertain Response of Ocean Biological Carbon Export in a Changing World." *Nature Geoscience* 15: 248–254. <https://doi.org/10.1038/s41561-022-00927-0>.
- Henson, S. A., R. Sanders, E. Madsen, P. J. Morris, F. Le Moigne, and G. D. Quartly. 2011. "A Reduced Estimate of the Strength of the Ocean's Biological Carbon Pump." *Geophysical Research Letters* 38, no. 4: L04606. <https://doi.org/10.1029/2011GL046735>.
- Huang, Y., A. J. Fassbender, J. S. Long, S. Johannessen, and M. Bernardi Bif. 2022. "Partitioning the Export of Distinct Biogenic Carbon Pools in the Northeast Pacific Ocean Using a Biogeochemical Profiling Float." *Global Biogeochemical Cycles* 36: e2021GB007178. <https://doi.org/10.1029/2021GB007178>.
- Johnson, L., D. A. Siegel, A. F. Thompson, et al. 2024. "Assessment of Oceanographic Conditions during the North Atlantic EXport Processes in the Ocean from RemoTe Sensing (EXPORTS) Field Campaign." *Progress in Oceanography*. 220: 103170. <https://doi.org/10.1016/j.pocean.2023.103170>.
- Laws, E. A. 1991. "Photosynthetic Quotients, New Production and Net Community Production in the Open Ocean." *Deep Sea Research Part A. Oceanographic Research Papers* 38: 143–167. [https://doi.org/10.1016/0198-0149\(91\)90059-O](https://doi.org/10.1016/0198-0149(91)90059-O).
- Laws, E. A., E. D'Sa, and P. Naik. 2011. "Simple Equations to Estimate Ratios of New or Export Production to Total Production from Satellite-Derived Estimates of Sea Surface Temperature and Primary Production: Equations for Ef Ratios." *Limnology and Oceanography: Methods* 9: 593–601. <https://doi.org/10.4319/lom.2011.9.593>.
- Laws, E. A., P. G. Falkowski, W. O. Smith Jr., H. Ducklow, and J. J. McCarthy. 2000. "Temperature Effects on Export Production in the Open Ocean." *Global Biogeochemical Cycles* 14: 1231–1246. <https://doi.org/10.1029/1999GB001229>.

- Martin, J. H., and S. E. Fitzwater. 1988. "Iron Deficiency Limits Phytoplankton Growth in the North-East Pacific Subarctic." *Nature* 331: 341–343. <https://doi.org/10.1038/331341a0>.
- McCave, I. N. 1975. "Vertical Flux of Particles in the Ocean." *Deep Sea Research and Oceanographic Abstracts* 22: 491–502. [https://doi.org/10.1016/0011-7471\(75\)90022-4](https://doi.org/10.1016/0011-7471(75)90022-4).
- McNair, H. M., F. Morison, J. R. Graff, T. A. Rynearson, and S. Menden-Deuer. 2021. "Microzooplankton Grazing Constrains Pathways of Carbon Export in the Subarctic North Pacific." *Limnology and Oceanography* 66: 2697–2711. <https://doi.org/10.1002/lno.11783>.
- Meyer, M. G., M. A. Brzezinski, M. Cohn, et al. 2023. "Primary Production Dynamics during the Decline Phase of the North Atlantic Annual Spring Bloom." *bioRxiv*. <https://doi.org/10.1101/2023.05.18.541304>.
- Meyer, M. G., W. Gong, and S. M. Kafrissen. 2022. "Phytoplankton Size-Class Contributions to New and Regenerated Production during the EXPORTS Northeast Pacific Ocean Field Deployment." *Elementa: Science of the Anthropocene* 10: 00068. <https://doi.org/10.1525/elementa.2021.00068>.
- Moreno, A. R., C. A. Garcia, A. A. Larkin, et al. 2020. "Latitudinal Gradient in the Respiration Quotient and the Implications for Ocean Oxygen Availability." *Proceedings of the National Academy of Sciences* 117: 22866–22872. <https://doi.org/10.1073/pnas.2004986117>.
- Nicholson, D., S. Emerson, and C. C. Eriksen. 2008. "Net Community Production in the Deep Euphotic Zone of the Subtropical North Pacific Gyre from Glider Surveys." *Limnology and Oceanography* 53: 2226–2236. https://doi.org/10.4319/lno.2008.53.5_part_2.2226.
- Nicholson, D. P., S. T. Wilson, S. C. Doney, and D. M. Karl. 2015. "Quantifying Subtropical North Pacific Gyre Mixed Layer Primary Productivity from Seaglider Observations of Diel Oxygen Cycles." *Geophysical Research Letters* 42: 4032–4039. <https://doi.org/10.1002/2015GL063065>.
- Organelli, E., G. Dall'Olmo, R. J. W. Brewin, G. A. Tarran, E. Boss, and A. Bricaud. 2018. "The Open-Ocean Missing Backscattering Is in the Structural Complexity of Particles." *Nature Communications* 9: 5439. <https://doi.org/10.1038/s41467-018-07814-6>.
- Palevsky, H. I., and S. C. Doney. 2018. "How Choice of Depth Horizon Influences the Estimated Spatial Patterns and Global Magnitude of Ocean Carbon Export Flux." *Geophysical Research Letters* 45: 4171–4179. <https://doi.org/10.1029/2017GL076498>.
- Sanders, R., S. A. Henson, M. Koski, et al. 2014. "The Biological Carbon Pump in the North Atlantic." *Progress in Oceanography* 129: 200–218. <https://doi.org/10.1016/j.pocean.2014.05.005>.
- Siegel, D. A., K. O. Buesseler, M. J. Behrenfeld, et al. 2016. "Prediction of the Export and Fate of Global Ocean Net Primary Production: The EXPORTS Science Plan." *Frontiers in Marine Science* 3: 22. <https://doi.org/10.3389/fmars.2016.00022>.
- Siegel, D. A., I. Cetinić, J. R. Graff, et al. 2021. "An Operational Overview of the EXport Processes in the Ocean from RemoTe Sensing (EXPORTS) Northeast Pacific Field Deployment." *Elementa: Science of the Anthropocene* 9, no. 1: 00107. <https://doi.org/10.1525/elementa.2020.00107>.
- Siegel, D. A., S. Maritorena, N. B. Nelson, D. A. Hansell, and M. Lorenzi-Kayser. 2002. "Global Distribution and Dynamics of Colored Dissolved and Detrital Organic Materials." *Journal of Geophysical Research, Oceans* 107: 21. <https://doi.org/10.1029/2001JC000965>.
- Sieracki, M. E., P. G. Verity, and D. K. Stoecker. 1993. "Plankton Community Response to Sequential Silicate and Nitrate Depletion during the 1989 North Atlantic Spring Bloom." *Deep-Sea Research Part II: Topical Studies in Oceanography* 40: 213–225. [https://doi.org/10.1016/0967-0645\(93\)90014-E](https://doi.org/10.1016/0967-0645(93)90014-E).
- Steeman Nielsen, E. 1952. "The Use of Radio-Active Carbon (C14) for Measuring Organic Production in the Sea." *ICES Journal of Marine Science* 18: 117–140. <https://doi.org/10.1093/icesjms/18.2.117>.
- Steinberg, D. K., C. A. Carlson, N. R. Bates, S. A. Goldthwait, L. P. Madin, and A. F. Michaels. 2000. "Zooplankton Vertical Migration and the Active Transport of Dissolved Organic and Inorganic Carbon in the Sargasso Sea." *Deep Sea Research Part A. Oceanographic Research Papers* 47: 137–158. [https://doi.org/10.1016/S0967-0637\(99\)00052-7](https://doi.org/10.1016/S0967-0637(99)00052-7).
- Stukel, M. R., H. Song, R. Goericke, and A. J. Miller. 2018. "The Role of Subduction and Gravitational Sinking in Particle Export, Carbon Sequestration, and the Remineralization Length Scale in the California Current Ecosystem." *Limnology and Oceanography* 63: 363–383. <https://doi.org/10.1002/lno.10636>.
- Takahashi, T., S. C. Sutherland, C. Sweeney, et al. 2002. "Global Sea-Air CO₂ Flux Based on Climatological Surface Ocean pCO₂, and Seasonal Biological and Temperature Effects." *Deep-Sea Research Part II: Topical Studies in Oceanography* 49: 1601–1622. [https://doi.org/10.1016/S0967-0645\(02\)00003-6](https://doi.org/10.1016/S0967-0645(02)00003-6).
- Van Haren, H., C. P. D. Brussaard, L. J. A. Gerringa, M. H. Van Manen, R. Middag, and R. Groenewegen. 2021. "Diapycnal Mixing across the Photoc Zone of the NE Atlantic." *Ocean Science* 17: 301–318. <https://doi.org/10.5194/os-17-301-2021>.
- Westberry, T. K., P. J. I. B. Williams, and M. J. Behrenfeld. 2012. "Global Net Community Production and the Putative Net Heterotrophy of the Oligotrophic Oceans." *Global Biogeochemical Cycles* 26: 1–17. <https://doi.org/10.1029/2011GB004094>.

Supporting Information

Additional Supporting Information may be found in the online version of this article.

Submitted 01 November 2023

Revised 12 June 2024

Accepted 01 February 2025